

Background

Study system: Mormyrids are a clade of African freshwater fish possessing both passive and active electrosensory systems¹, as well as large brains for their body size².

- Electric organs produce electric organ discharges (EODs) used for communication and navigation¹.
- EOD production is hypothesized to be an energetically costly trait, based on research done on other clades of electric fish².
- However, mormyrids are extremely encephalized, unlike other clades of electric fish³.

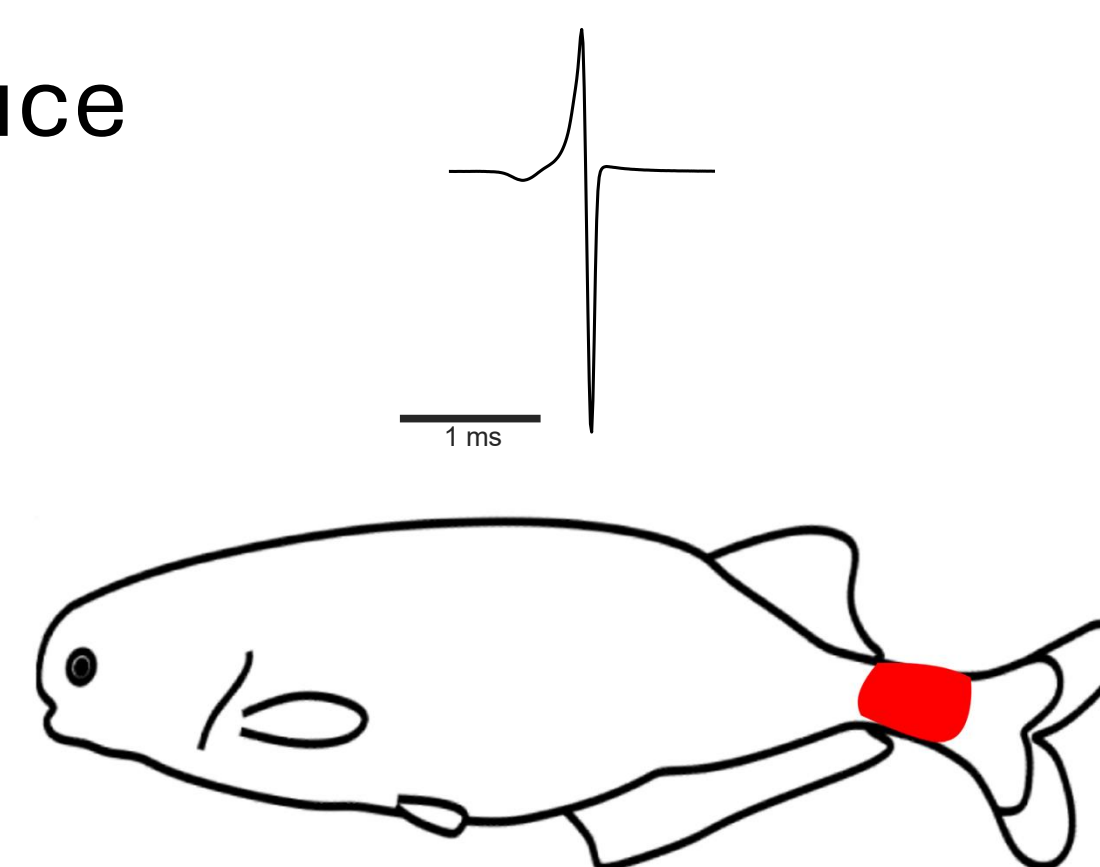


Figure 1. *Brienomyrus brachyistius* and example EOD. Red portion is caudal peduncle, where electric organ is located.

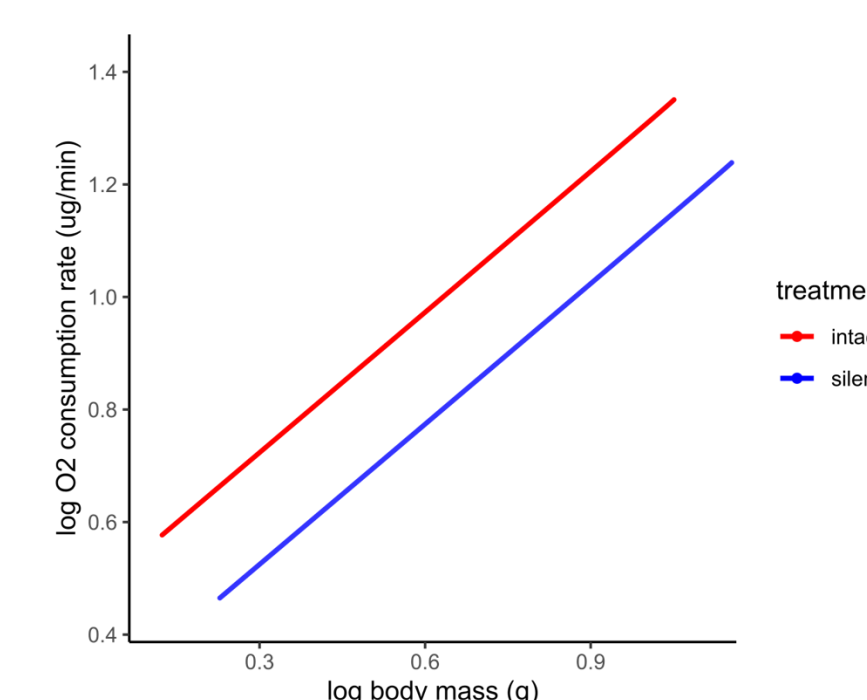


Figure 2. Prediction of oxygen consumption rates for intact and silent fish.

What is the cost of EOD production?

Methods – Silencing Surgery

To silence EOD production, we transected the spinal cord directly anterior to the electric organ. This method disrupts EOD production without disrupting motor function.

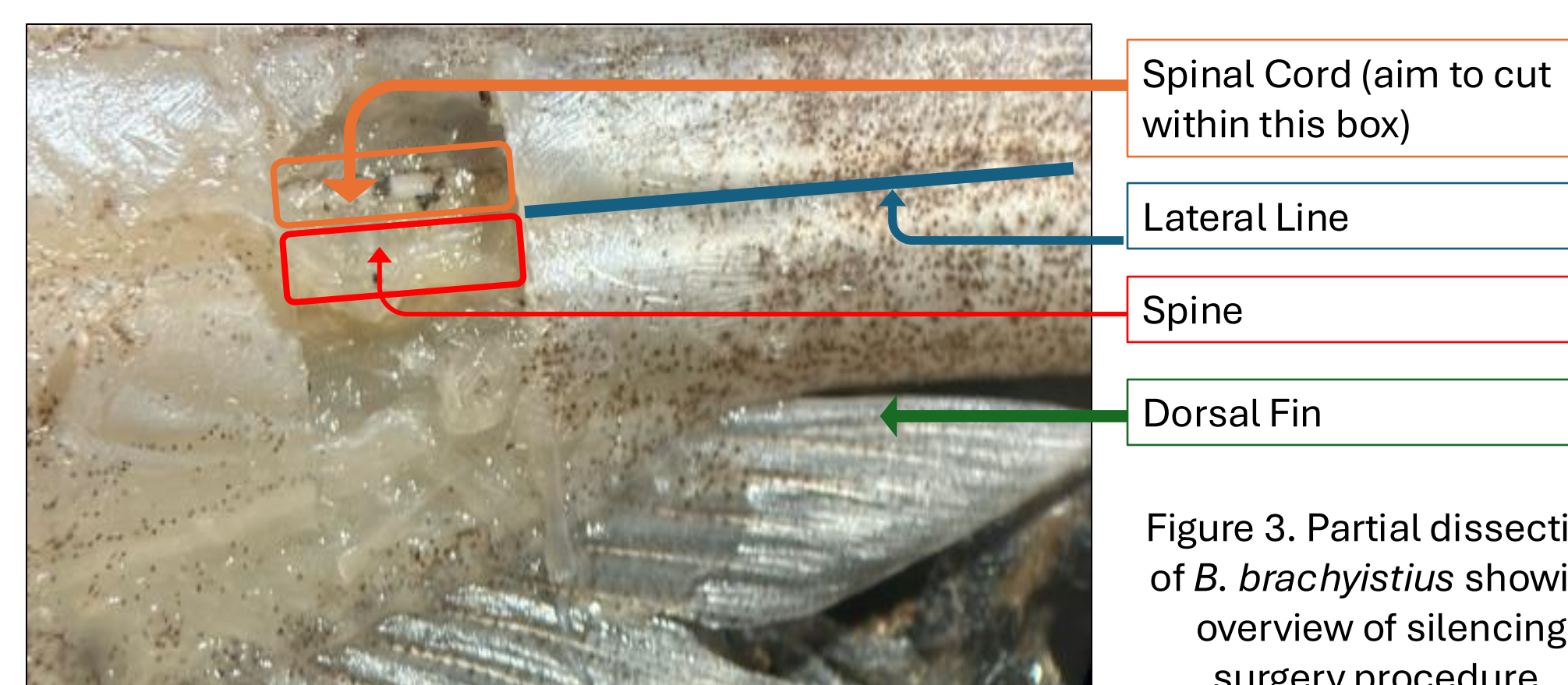


Figure 3. Partial dissection of *B. brachyistius* showing overview of silencing surgery procedure.

Methods - Respirometry

We used whole animal respirometry in *Brienomyrus brachyistius* to compare oxygen consumption rates between fish with intact and surgically silenced EOD production.

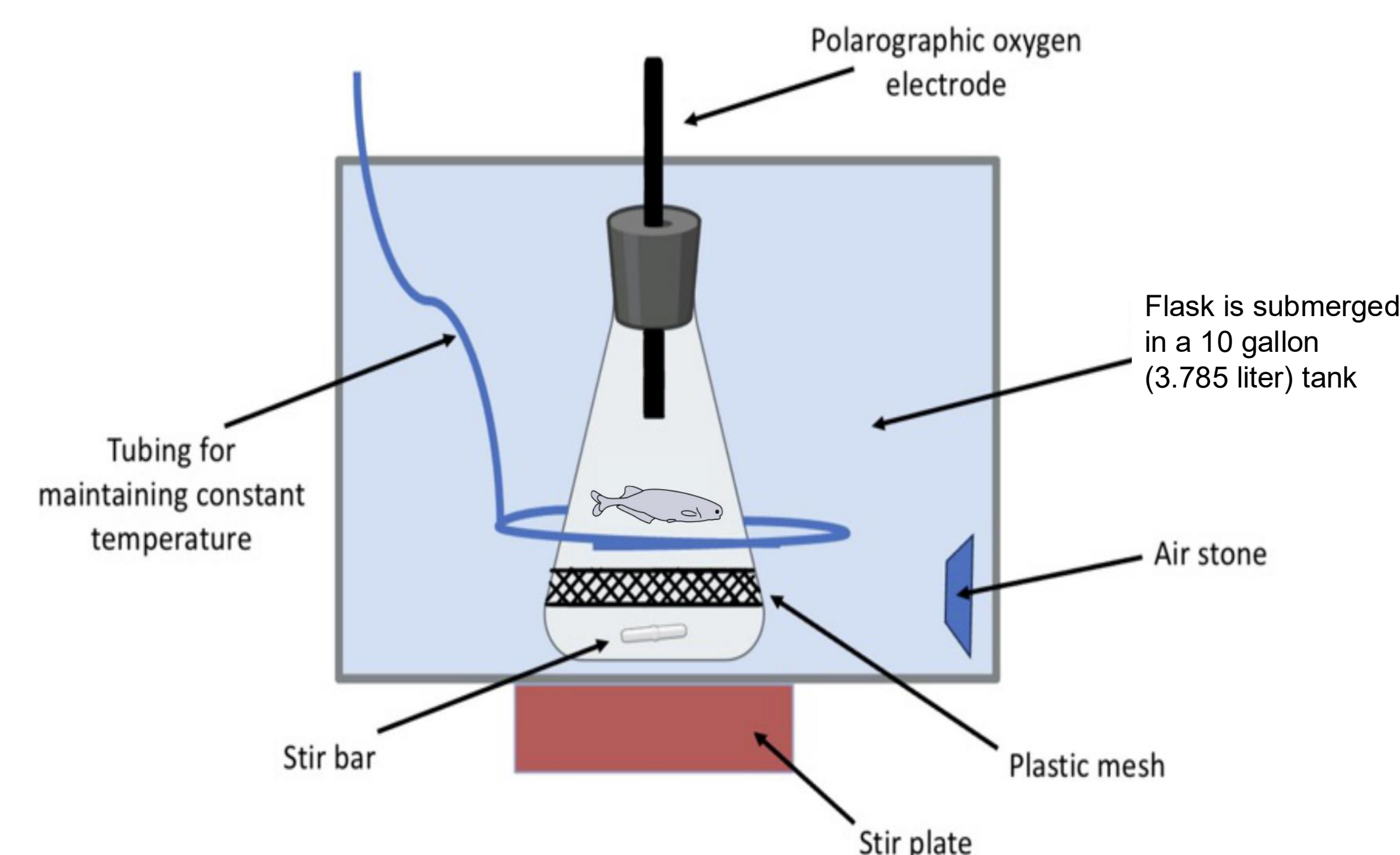


Figure 4. Schematic of whole animal respirometry setup.

Results

After controlling for body mass, we found no statistically significant difference in oxygen consumption between intact and surgically silenced fish (ANCOVA: $F(1,32) = 0.502$, $p = 0.484$). Body mass was a significant covariate ($F(1,32) = 102.538$, $p < 0.001$).

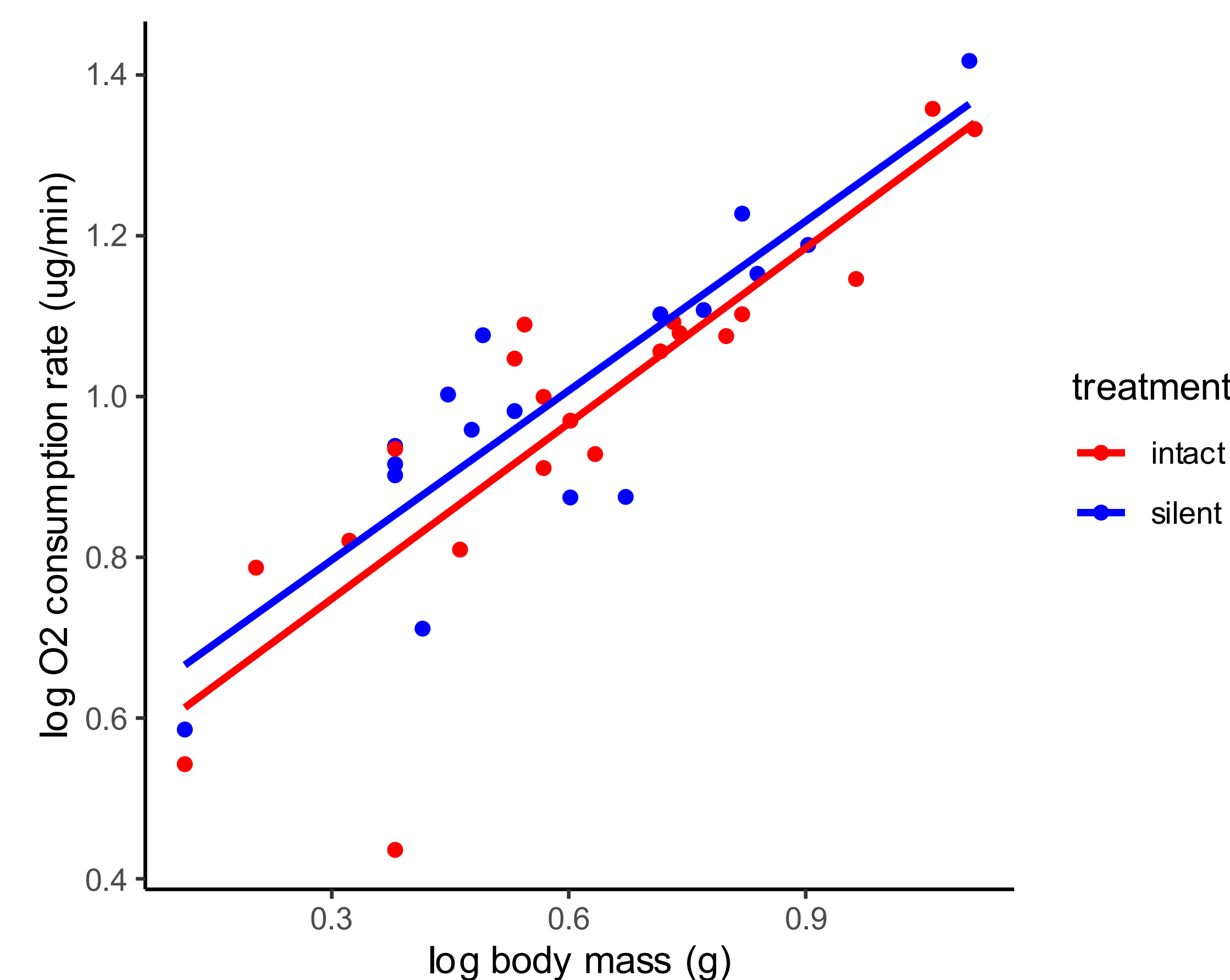


Figure 5. There is no significant difference between intact and silent oxygen consumption rates.

Conclusions

- Overall metabolic rate (total energy expenditure) is not affected by the use or absence of EODs.
- Metabolic rate within this species may be a physiologically constrained trait⁴.

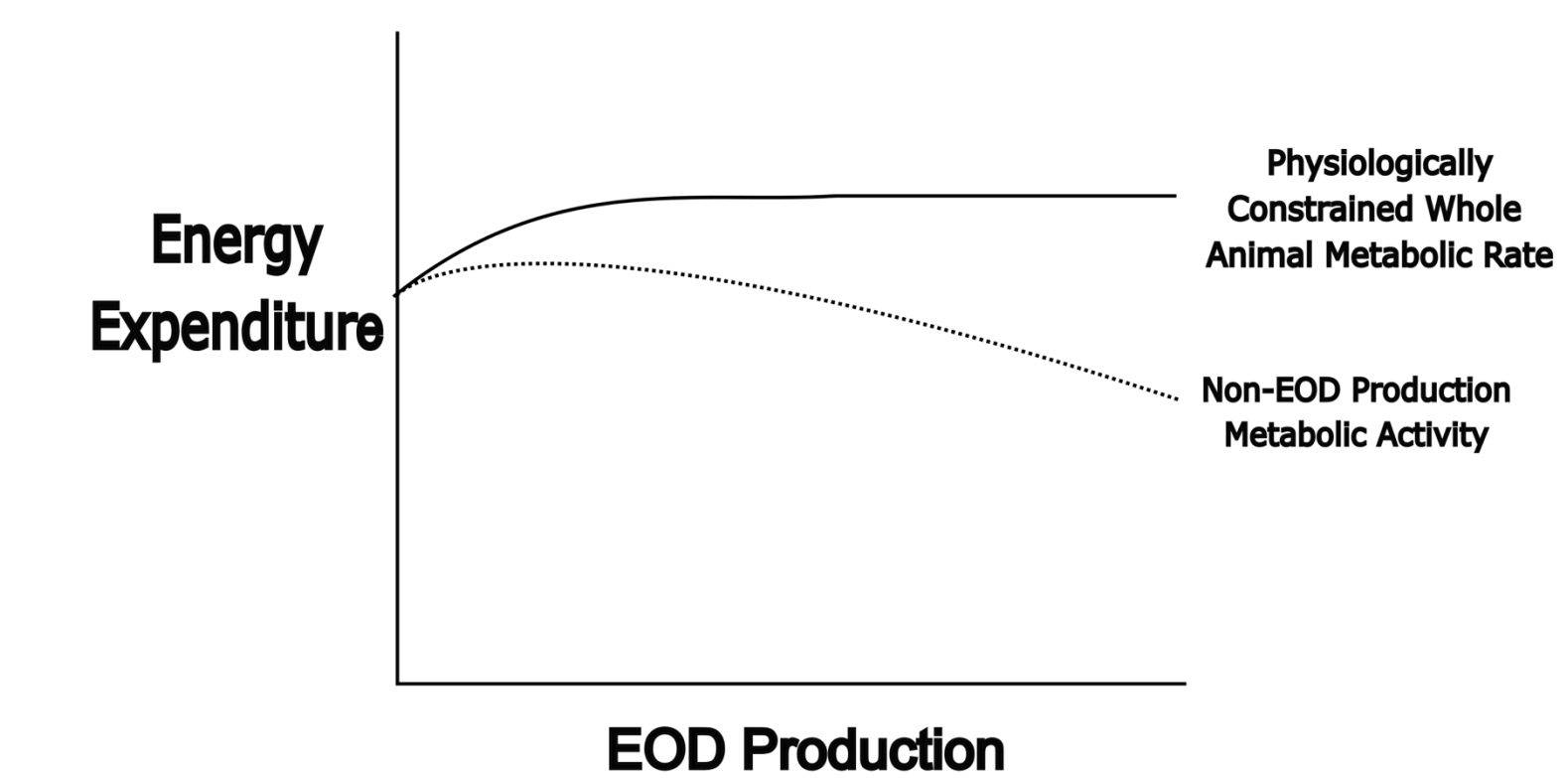


Figure 6. Schematic of Total Energy Expenditure (TEE) modeled as a physiologically constrained trait. Total energy expenditure does not change with increasing physical activity. Non-physical activity energy expenditure declines with increasing physical activity. Figure modified from Pontzer 2015⁴.

Future Directions

- In mormyrids, is the physiologically constrained nature of the metabolic rate unique to *B. brachyistius*, the species with the smallest relative brain size? Repeat experiments with different species of different relative brain sizes.
- Explore if energy is being reallocated to growth in the absence of EOD production.
- What are the fine grain energetic costs of EOD production? Determine ATP usage per EOD in an electric organ of given size.

References & Acknowledgements

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